

Fibonacci sequence in music

Gilded Runner
1+1+2+3+5+8+13+21

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Allegro Risoluto ♩ = 94

The musical score for 'Gilded Runner' is presented in five systems, each corresponding to a Fibonacci number in the sequence 1, 1, 2, 3, 5, 8, 13, 21. The tempo is marked 'Allegro Risoluto' with a quarter note equal to 94 beats. The key signature is B-flat major (two flats). The first system (measures 1-6) is marked 'mp' (mezzo-piano). The second system (measures 7-13) and fourth system (measures 9-21) are marked 'mf' (mezzo-forte). The third system (measures 14-21) is marked 'f' (forte). The fifth system (measures 22-42) is marked 'ff' (fortissimo). The measure counts for each system are: 6, 7, 13, 18, and 21 measures respectively, which correspond to the Fibonacci sequence values.

The Fibonacci sequence is explained by the function $F(n) = F(n-1) + F(n-2)$.

$n = 0, 1, 2, 3, 4, 5, 6, 7, 8$

$F(n) = 0, 1, 1, 2, 3, 5, 8, 13, 21$

As it goes:

$$0 + 1 = 1$$

$$1 + 1 = 2$$

$$1 + 2 = 3$$

$$2 + 3 = 5$$

$$3 + 5 = 8$$

$$5 + 8 = 13$$

$$8 + 13 = 21 \dots$$

The same sequence can be seen in the song Gilded Runner from Genshin Impact. The time signatures go in the order 1/8, 1/8, 2/8, 3/8, 5/8, 8/8, 13/8, 21/8. This shows the practice and use of the Fibonacci sequence in a practical sense.

The top number in a time signature is how many beats a minute can be in a measure. The bottom number is how long a beat/note can be. So in /8 time the notes would be 1/8th long.



The most common time signature is 4/4. So 4 beat per measure, and the beats/notes are 1/4 or a quarter long.



Sources

https://musescore.com/user/18215371/scores/12367261?srltid=AU7gw4Xwr5ZnfMjontXBUXH9-rObEtFkDCFlGQJiDAs_rqXtCnWuTW3A